

HAT-P-19b

R-1569

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_220810 · created 2026-08-01T22:24:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459801.8621 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.052
Transit depth (Rp/Rs)^2	3.14 +/- 1.86 [%]
Orbital Inclination (inc)	86.59 +/- 1.54
Airmass coefficient 1 (a1)	0.24 +/- 0.16
Airmass coefficient 2 (a2)	1.24 +/- 0.38
Scatter in the residuals of the lightcurve fit is	76.35 %
Best Comparison Star	None
Optimal Aperture	3.41
Optimal Annulus	4.26
Transit Duration (day)	0.078 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.052 vs 0.1418 (deviation 0.68σ)
Duration fit vs published	0.078 d vs 0.1196 d (deviation 1.01σ)
Mid-time O–C	9.6 min
Depth SNR	3.4
Residual scatter	76.35 %

Light curve

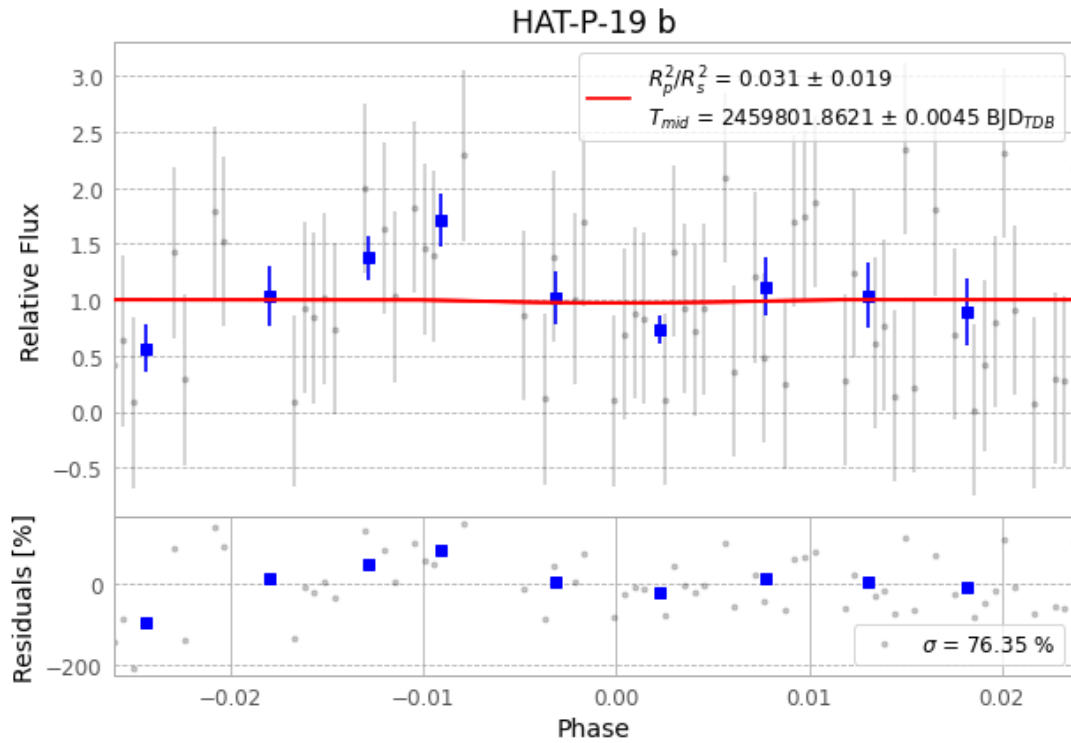


Figure 1 — FinalLightCurve_HAT-P-19 b_2022-08-09.png (SHA-256 b76e7d0e3ecb8be5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [153, 46], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cec78b6e7c97963f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), HATP-19220810060613.FITS → HATP-19220810105413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220810.log (db1d69e921a2...), FinalLightCurve_HAT-P-19 b_2022-08-09.png (b76e7d0e3ecb...), FinalLightCurve_HAT-P-19 b_2022-08-09.csv (d9c62c9ee969...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:24Z.